Promo Finder: finding promos in a microdrama screenplay (Problem 1)

Input: *I Married a Broken Soldier. He was a Duke*, 50 episodes

Why Problem 1

It's a real bottleneck with a clear owner: an editor watches every episode to pick promos, one show at a time, while new shows ship every week. The input is fixed and the output is checkable (timestamps, a 20–60 s length, a hook and a cliffhanger), so I can measure where the tool is right and wrong instead of arguing about it. And the AI is only useful for one part of it (judging drama), which made "what to give the AI and what to keep as code" a concrete design question.

Assumptions (where the brief was silent)

- **Script timing = edit timing.** Timestamps, the 20–60 s length and "hook in the first 3 s" are measured on the script. The edited footage may differ slightly.
- **A promo is 1–3 pieces of one episode.** Cross-episode promos are out of scope: joining moments across episodes risks clips that only make sense with backstory, so I prioritised single-scene promos a new viewer can follow.
- **The viewer knows nothing**, but not knowing names or backstory is normal for a promo; what matters is whether the conflict (who is wronging whom, what's at stake) reads.
- **Free first, model swappable.** The default is Gemini's free tier. Its daily limit stopped my full run after ~19 episodes, so the submitted run uses gpt-5-mini (paid, ~$0.53 per show); the code is identical, the model is one setting.
- **Confidentiality.** The repo is public and holds code only; the scripts and outputs (which quote script lines) are shared via a link, not published.

How it works: what the AI does and what code does

```
scripts → parse into numbered, timed blocks (code) → per episode: AI proposes ≤3 promos as block IDs
        → check + fix each proposal (code) → one repair round (AI) → remove duplicates (code)
        → blind check of the top 30 (AI) → best 15 picked and ordered (AI) → variety rule (code)
```

Decision	Why
The model answers with **block IDs** (`B027`), never timestamps; code looks the times up	An LLM writes plausible-looking wrong numbers. An ID that doesn't exist is caught instantly; a wrong timestamp isn't.
Code enforces the brief's rules: 20–60 s, hook within 3 s, cut on the cliffhanger, never mid-line	Rules should never depend on the model following instructions. Obvious fixes are automatic (cut the lead-in, trim the tail); anything needing judgement goes back to the model **once** with the exact error.
Duplicates = same hook, same cliffhanger, or >80% shared footage	My first rule (50% overlap) deleted a genuinely different ad; overlapping clips with different hooks and endings are different promos.
A **blind check**: a fresh call sees only the clip's lines, judged as a stranger	The picking model has read the whole episode, so its own "stands alone" score is optimistic.
AI picks and orders the best 15 from the top 30, then a **variety rule in code** (≤2 promos from any 5-episode window in the top 10)	In my first full run the ranking model, though told to vary the order, put 5 clips from one storyline in its top 15.
Every model sits behind one interface; provider, model and keys are in `.env`	Kuku can plug in its own model with `--model custom` and a URL; no code change.

One promo as the tool outputs it (from `report.md`; every promo also lists auto-fixes and warnings):

```
P005 · Garden Ambush — Betrayal · 38 s · rank 9
Segment:      Episode 16, 00:14.30 → 00:52.80
Hook (0–3 s): A thug lunges with a club; Alexander catches the iron pipe mid-swing with his bare hand.
              script: CAMERA: Medium action shot as the Thug with Cap lunges forward with his weapon raised high.
Cliffhanger:  THUG WITH CAP (Screaming, Panicked): "Harrington! Gerald Harrington paid us!"
Why:          Instant physical hook, then a clear, high-stakes betrayal reveal.
Scores:       hook 9 · cliffhanger 9 · standalone 7 (picker) / 9 (blind)
```

Scale and cost

- **Per show (50 episodes):** 94 model calls (50 find, 13 repair, 30 blind, 1 ranking), about 261k input and 233k output tokens: **~$0.53** at assumed gpt-5-mini prices ($0.25 / $2 per million tokens), ~10 minutes with 4 parallel workers. Ten shows a week is about $5.
- **Human time:** an editor reviews the top 15 clips (~20 min) instead of watching ~100 minutes of episodes. The editor stays in the loop for the final pick.
- **Levers:** leaving camera lines out halves input tokens; any cheaper model is one setting; a free keyword baseline runs with no model at all. Answers are cached, so reruns and resumed runs cost nothing.
- **Limit:** free tiers don't cover volume (Gemini's daily cap is below one show), so production needs a paid tier or Kuku's own endpoint.

How I know it's good, and where it fails

Rules hold. 104 promos, all 20–60 s (avg 37.5 s). Code auto-fixed 24 and the repair round rescued 17; 36 candidates were dropped, each with a recorded reason (19 wrong length, one of them after repair; 17 duplicates).

Against a free baseline. A keyword search (dramatic words + emotion tags) found the same moment for 74% of the AI's picks, and for 11 of its top 15. So the AI's value is less in *spotting* drama than in where to cut, what reads to a stranger, and ranking. The baseline's typical error: it picked "He strikes flint and steel" (lighting a fire) because of the word "strikes".

Hand review of the top 15 (AI-drafted first pass, checked and confirmed by me): **5 post, 7 post with an edit, 3 don't post.**

The blind check was the most instructive failure. I changed only the wording of its scoring rubric:

Rubric	Blind scores	Effect
v1: as first written	mostly 8 (21 of 30), none below 5	demoted nothing
v2: "an unknown character → score ≤ 6"	45 of 72 scored 5	the show's premise clip (jilted at the altar) fell from #1 to #30
v3: "only conflict clarity counts"	8–10 for all 30 (avg 9.0)	lenient again

Against my verdicts it barely separates good from bad: average blind score **9.2** for "post", **9.3** for "edit", **8.7** for "don't post"; two don't-posts scored 9. **Conclusion:** a single LLM judge's number swings with rubric wording and doesn't match human judgement. Its written notes ("who set the fire, and why?") are useful to a reviewer; its score isn't a filter.

Where it picks badly (from the top 15):

- **Same beat twice:** #1 and #7 both end with a red-hot iron at Juliana's eye (adjacent episodes, so the variety rule allows both). It counts episodes, not story beats.
- **Late reveals cluster:** #3, #5 and #6 are versions of the Duke reveal (E42–45); the ±2-episode window is narrower than that storyline.
- **Payoff instead of cliffhanger:** #3 ends with the guards siding with Alexander, which resolves the moment.
- **Hooks that need backstory:** #6 ("the lie you built your life upon"); #15 ("Arsenic." means nothing at second 1).
- **No content-policy check:** #1 is graphic violence; #13 is a sexual scene with a priest, and the villain's subplot, not the heroine's story.
- **The premise clip ranks #21.** "Abandoned at the Altar" is arguably the best first promo for new viewers; the ranking model left it out of its top 15.
- **Text only:** 5 of the top 15 open on a camera direction, so whether the hook lands depends on footage the tool can't see.

What I'd post first, and why

The tool's order, corrected by my review: clips a stranger gets in seconds, spread across the show, with anything that needs a policy check held back.

#	Promo (tool rank)	Segment	Why first
1	P017 Abandoned at the Altar (#21)	E01 00:39.10–01:03.00 (24 s)	The show's premise: jilted at the altar, sister gloating, "I changed my mind." Reads with zero context. My one override of the tool's order.
2	P005 Garden Ambush (#9)	E16 00:14.30–00:52.80 (38 s)	Club caught bare-handed, then "Gerald Harrington paid us!": action plus a betrayal that works without knowing any names.
3	P030 The Cathedral Blast (#11)	E41 00:57.80–01:36.09 (38 s)	Defiant line, an explosion, armed figures in the smoke, open ending. The best of the cathedral clips.
4	P025 Door-Kicked Rescue (#10)	E18 00:17.80–01:02.50 (45 s)	Crash, fire and rescue in the first seconds. Check the opening whisper in the footage.
5	P024 Plant Smash (#14)	E15 00:03.60–00:34.00 + 01:03.20–01:26.00 (53 s)	A woman bullied, then the hero arrives; the two-piece cut skips the slow middle.
6	P002 The Branding Threat (#1)	E32 01:59.50–02:34.38 (35 s)	The strongest hook and cliffhanger in the run, but graphic violence: post only after a platform-policy check.

What I'd do next

- **Calibrate the blind judge** on ~50 hand-labelled clips, or switch it to pairwise "which clip is clearer?", until its scores agree with an editor's.
- **Variety by story beat:** have the ranking model tag each clip's beat (altar betrayal, Duke reveal…) and cap beats, not episodes, in the top 10.
- **Policy check and feedback:** flag violent or sexual clips before posting, and feed real promo performance (watch-through, installs) back into the score weights.

AI tools I used, and one place they got it wrong

I built this with **Claude Code**: I set the requirements and the design decisions above, it wrote most of the code, and I reviewed each file and tested every change on real runs. **One place it got it wrong:** its first duplicate rule treated any two clips sharing 50% of their footage as the same promo, and on episode 1 it deleted a genuinely different ad ("abandoned at the altar" vs "the sister's betrayal": different hooks, different endings). I only caught it by reading the first run's output; the rule now compares hooks and cliffhangers. More generally, AI-written rubrics need measuring: the v2 blind-check rubric over-penalised promos for not explaining backstory, which promos never do.